

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Account Creation & Session Retention	Sub-Task 1.1: Build register component with match confirmations and username validations. Sub-Task 1.2: Implement bcryptjs hashing (10 salt rounds) on backend database model. Sub-Task 1.3: Setup Express endpoints to issue signed JWTs with 24-hour expirations. Sub-Task 1.4: Implement browser localStorage management routines on the frontend.
FR-2	Catalog Filter Engine	Sub-Task 2.1: Code regular expression database query parsers for text strings. Sub-Task 2.2: Build multi-select category checkboxes that update search queries in the URL. Sub-Task 2.3: Add sorting queries for prices (low-to-high, high-to-low) and discount margins. Sub-Task 2.4: Implement real-time client search input debounce limits to throttle server calls.
FR-3	Interactive Cart Management	Sub-Task 3.1: Setup database operations to load user-scoped shopping baskets. Sub-Task 3.2: Connect addition controls that bundle matching sizes into single records. Sub-Task 3.3: Add click handlers (+ / -) that update unit counts in the database. Sub-Task 3.4: Build receipt calculators summarizing retail totals, active promotions, and delivery offsets.
FR-4	Form Checkout & Shipping Validation	Sub-Task 4.1: Build inputs capturing recipient names, mobile lines, pincodes, and street addresses. Sub-Task 4.2: Code formatting validators verifying 10-digit mobile structures. Sub-Task 4.3: Build processing controls to submit records to the orders database table. Sub-Task 4.4: Trigger post-purchase cleanup processes to wipe active carts.
FR-5	Unified Admin Dashboard	Sub-Task 5.1: Set up layout state listeners that apply the .admin-theme styling overlay. Sub-Task 5.2: Build aggregation endpoints that calculate total user registers, product inventories, and sales logs.

		Sub-Task 5.3: Build forms allowing administrative users to edit catalog details. Sub-Task 5.4: Create status toggles for orders (order placed, in transit, delivered, cancelled).
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Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Responsive UI across mobile, tablet, and desktop views down to 320px screen widths; utilizes the fast-loading Outfit typography from Google Fonts; interactive UI components feature hover highlights with CSS transition durations set to 200ms.
NFR-2	Security	Password hashing verified through bcryptjs encryption; API validation layers enforce route security via HTTP headers (Authorization: Bearer <token>); administrative endpoints return 403 Forbidden for non-admin accounts.
NFR-3	Reliability	Client API connections handle network failures gracefully using session recovery routines; local DB proxies fall back automatically during MongoDB Atlas service timeouts, ensuring offline sandbox stability.
NFR-4	Performance	Keep React build bundles under 250KB compressed for fast initial paint times on mobile networks; API query times for catalog routes remain below 200ms; supports image lazy loading on the catalog search grids.
NFR-5	Availability	Target a 99.9% application uptime by hosting on load-balanced instances with health checks.
NFR-6	Scalability	Decoupled server-client routing allows scaling hosting resources for backend microservices and frontend static hosts independently.